

Kapil Kumar Meena

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Interests Travel behaviour and discrete choice modelling; air-pollution and heat exposure in daily travel; information, electric mobility and decision support; machine learning and behaviourally grounded AI for transportation; safety in heterogeneous traffic; transport policy.

Record 1 published patent, 9 journal articles, 7 manuscripts under review and 17 refereed conference papers. Google Scholar: 200+ citations, h-index 6 (August 2026). Reviewer for 8 journals.

APPOINTMENTS

2026– **Postdoctoral Researcher** Los Angeles, CA
HUMAN Lab, University of California, Los Angeles
With Prof. Youngseo Kim. Structured discrete choice estimation (PD-MUSE, under review at *Transportation Research Part C*); heat, exposure and daily activity and travel (HEAT).

2024 **Visiting Research Fellow** Leeds, UK
Institute for Transport Studies, University of Leeds
International Research Exchange, jointly funded by the University of Leeds and IIT Kharagpur.

2021–2022 **Research Intern, Electric Mobility** Remote
WRI India
Inhaled-dose comparison for commuters across electric and internal-combustion fleets; recommendations for electric mobility in tier-2 cities.

EDUCATION

2022–2026 **Ph.D., Transportation Engineering** Kharagpur, India
Indian Institute of Technology Kharagpur
RCG School of Infrastructure Design and Management. Advisor: Prof. Arkopal Kishore Goswami. Thesis: *Assessing impacts of perceived air pollution exposure on urban travel behaviour*.

2019–2021 **M.Tech, Transportation Engineering** Roorkee, India
Indian Institute of Technology Roorkee
Advisor: Prof. Amit Agarwal. Thesis: *Air quality perception of commuters in Delhi*.

2015–2019 **B.Tech (Honours), Civil Engineering** Kota, India
Rajasthan Technical University
Top 5% of class.

PATENT

2026 **Personalised dynamic route planning system for sustainable urban mobility**
Indian Patent Application No. 202631018379, published 31 July 2026
Inventors: **Meena, K. K.**, Goswami, A. K.
The routing system behind the DRUM app (see J8 and Research software). Featured in *The Hindu*, June 2025.

MEDIA COVERAGE

Jun 2025 ***The Hindu*: “New app helps commuters pick ‘greener’ routes on road”**
By Ashmita Gupta, 8 June 2025. A feature on the DRUM routing app and its results for Delhi.
[Read online](#)

PUBLICATIONS

Journal articles

- J9 **Meena, K. K.**, Goswami, A. K. (2026). Not all travellers think alike: Segmenting travel behaviour under air pollution exposure using a hybrid latent class and discrete choice approach. *Transportation Research Record*. Accepted. doi:10.1177/03611981261429472
- J8 **Meena, K. K.**, Singh, A. K., Goswami, A. K. (2025). Dynamic route planning for urban green mobility: Development of a web application offering sustainable route options to commuters. *Transportation Research Record*. doi:10.1177/03611981251331011
- J7 Sumbhate, A., **Meena, K. K.**, Goswami, A. K. (2025). Assessing air pollution exposure to school children in different modes of transport while commuting to school: A case of Kharagpur, India. *Journal of the Eastern Asia Society for Transportation Studies*, 16. doi:10.11175/easts.16.PP3879
- J6 Manoj, B. S., **Meena, K. K.**, Panchal, H., Sharma, G., Goswami, A. K. (2025). Modeling bicycle choice behavior and its potential health impact: Case of first/last mile access to suburban rail. *International Journal of Sustainable Transportation*. doi:10.1080/15568318.2025.2572818
- J5 Manoj, B. S., **Meena, K. K.**, Goswami, A. K. (2025). A prioritization framework to identify key attributes of transit-oriented development (TOD) using a multi-criteria decision-making (MCDM) approach: An Indian context. *Sustainable Transport and Livability*, 2. doi:10.1080/29941849.2025.2516475
- J4 **Meena, K. K.**, Goswami, A. K. (2024). A review of air pollution exposure impacts on travel behaviour and way forward. *Transport Policy*, 154, 48–60. doi:10.1016/j.tranpol.2024.05.024
- J3 **Meena, K. K.**, Bairwa, D., Agarwal, A. (2024). A machine learning approach for unraveling the influence of air quality awareness on travel behavior. *Decision Analytics Journal*, 11, 100459. doi:10.1016/j.dajour.2024.100459
- J2 **Meena, K. K.**, Singh, V., Agarwal, A. (2023). Perception of commuters towards air quality in Delhi. *Journal of Transport & Health*, 31, 101643. doi:10.1016/j.jth.2023.101643
- J1 Singh, V., **Meena, K. K.**, Agarwal, A. (2021). Travellers' exposure to air pollution: A systematic review and future directions. *Urban Climate*, 38, 100901. doi:10.1016/j.uclim.2021.100901

Manuscripts under review

- R7 **Meena, K. K.**, Sesidhar, S. V. (2026). How charging information shapes choice, reliability, and trust in public EV charging. *Transportation Research Part D: Transport and Environment*. Second revision.
- R6 **Meena, K. K.**, Goswami, A. K. (2026). Do commuters reroute for cleaner air? Seasonal salience in route choice behaviour from a two-wave panel in Kolkata, India. *Travel Behaviour and Society*. First revision. Preprint: SSRN 6904230
- R5 **Meena, K. K.**, Tang, S., Kim, Y. (2026). PD-MUSE: Primal-dual maximum-utility structured estimator for discrete choice models. *Transportation Research Part C: Emerging Technologies*. Under review.
- R4 Sesidhar, S. V., **Meena, K. K.** (2026). Pricing shade in urban mobility: Behavioural evidence and optimal allocation for heat-resilient access in Delhi. *Transport Policy*. Under review.
- R3 Basu, S., Guha Majumdar, S., **Meena, K. K.**, Sesidhar, S. V. (2026). Are closer encounters really more dangerous? Evidence from conflict screening in heterogeneous traffic. *IATSS Research*. Under review.
- R2 **Meena, K. K.**, Goswami, A. K. (2026). Beyond time and cost: Seasonal air-quality preferences in travel mode choice from a two-wave commuter panel. *Transportation Research Record*. Under review. Accepted for presentation at TRB 2027 (C17).
- R1 **Meena, K. K.**, Goswami, A. K. (2025). Beyond choice modelling: Bridging econometric theory and deep learning for robust mode choice prediction. *Transportation Research Part A: Policy and Practice*. Under review.

Refereed conference papers

- C17 **Meena, K. K.**, Goswami, A. K. (2027). Beyond time and cost: Seasonal air-quality preferences in travel mode choice from a two-wave commuter panel. *106th Annual Meeting of the Transportation*

Research Board, Washington, DC. Accepted for presentation.

- C16 Basu, S., Guha Majumdar, S., **Meena, K. K.**, Sesidhar, S. V. (2027). Can one traffic image support conflict screening? A validity audit of a heterogeneous graph and grounded LLM framework in unstructured Indian traffic. *106th Annual Meeting of the Transportation Research Board, Washington, DC*. Accepted for presentation.
- C15 **Meena, K. K.**, Goswami, A. K. (2026). Not all travellers think alike: Segmenting travel behaviour under air pollution exposure. *105th Annual Meeting of the Transportation Research Board, Washington, DC*.
- C14 Gupta, C., Amitabh, A., **Meena, K. K.**, Goswami, A. K. (2026). A hybrid geostatistical and deep learning framework for urban pollutant concentration prediction from sparse data. *18th International Conference on Communication Systems & Networks (COMSNETS), IEEE*. doi:10.1109/COMSNETS67989.2026.11418296
- C13 Joshi, V., **Meena, K. K.**, Goswami, A. K. (2026). GeoNBeats: Unified spatio-temporal neural basis expansion for air quality estimation in sparse sensor networks. *IEEE International Conference on AI Engineering and Innovation (AIEI)*. doi:10.1109/AIEI69164.2026.11496657
- C12 Singh, A., **Meena, K. K.**, Sharma, G., Goswami, A. K., Mishra, S. (2026). Developing an integrated walkability score using image-based feature extraction and user preferences. *105th Annual Meeting of the Transportation Research Board, Washington, DC*. Also accepted at WCTR 2026, France.
- C11 **Meena, K. K.**, Singh, A. K., Goswami, A. K. (2025). Dynamic route planning for urban green mobility. *7th Conference of the Transportation Research Group of India (CTRG), SVNIT Surat*.
- C10 Sumbhate, A., **Meena, K. K.**, Goswami, A. K. (2025). Assessing the air pollution exposure to school children in different modes of transport while commuting to school. *Eastern Asia Society for Transportation Studies (EASTS) Conference, Jakarta*.
- C9 Manoj, B. S., **Meena, K. K.**, Goswami, A. K. (2025). A prioritization framework to identify key attributes of transit-oriented development (TOD). *1st World Symposium on Sustainable Transport and Livability (WSSTL), IISc Bengaluru*.
- C8 Kodukulla, R., **Meena, K. K.**, Sharma, G., Goswami, A. K. (2025). Accessibility assessment of urban public transit to key facilities through spatial analysis: A case study of Delhi. *Transportation Infrastructure Projects: Conception to Execution (TIPCE), IIT Roorkee*.
- C7 Dasgupta, S., **Meena, K. K.**, Majumdar, D., Goswami, A. K. (2025). Air pollution exposure among Kolkata's auto-rickshaw drivers: PM variability, health risks and predictive modelling. *Advances in Energy Research (AEEE), India*.
- C6 **Meena, K. K.**, Taneja, R., Agarwal, A. (2024). Impact of air pollution on informed decision-making for choice of a travel mode. *16th International Conference on Communication Systems & Networks (COMSNETS), IEEE*. doi:10.1109/COMSNETS59351.2024.10427003
- C5 Sumbhate, A., **Meena, K. K.**, Goswami, A. K. (2024). Breathable modes to school: Assessing the air pollution exposure of travel choices for school children in urban environments. *52nd Urban Affairs Association (UAA) Annual Meeting, Nashville*.
- C4 Mohanty, P., **Meena, K. K.**, Goswami, A. K. (2024). Analysing user behaviour along dedicated bicycle facilities in an urban environment. *52nd Urban Affairs Association (UAA) Annual Meeting, Nashville*.
- C3 Manoj, B. S., **Meena, K. K.**, Panchal, H., Sharma, G., Goswami, A. K. (2024). Assessing the willingness to bicycle for the first mile to the Mumbai suburban rail. *17th International Conference on Travel Behaviour Research (IATBR), Vienna*.
- C2 **Meena, K. K.**, Goswami, A. K. (2023). A review of air pollution exposure impacts on travel behaviour and way forward. *16th World Conference on Transport Research (WCTR), Montréal*.
- C1 **Meena, K. K.**, Kumar, R., Goswami, A. K. (2022). On-road pollution exposure in multiple transport micro-environments: A case study of tier-2 and tier-3 cities in India. *14th International Conference on Transportation Planning and Implementation Methodologies for Developing Countries (TPMDC), IIT Bombay*.

AWARDS, FELLOWSHIPS AND GRANTS

2026	Institute Travel Grant (full funding), TRB Annual Meeting, IIT Kharagpur
2024	International Research Exchange Grant, University of Leeds and IIT Kharagpur
2024	Institute Travel Grant (partial funding), IATBR, Vienna, IIT Kharagpur
2024	Best Presentation Award, Research Scholar Day, RCGSIDM, IIT Kharagpur
2023	Best Poster Presentation, Cyber-Physical Systems Summit (CyPhySS), IISc Bengaluru
2019–2026	MHRD Fellowship, Ministry of Education, Government of India
2019	GATE Civil Engineering, All India Rank 559
	Regional winner, National Children's Science Congress, Department of Science & Technology, Government of India

INVITED TALKS

2022	Transportation data analysis using real-world mobility data. NIT Calicut.
2022	Data visualisation and decision support using Tableau. Swastik Edustart.
2020	Quality control and performance evaluation of road infrastructure. IIT Roorkee.
2019	Practical aspects of concrete mix design and quality assurance. Wonder Cement Ltd.

TEACHING

Interests	Core courses in transportation engineering, traffic engineering and transportation planning; electives in travel demand and discrete choice analysis, transportation data analytics and machine learning, and transport, environment and health.
2019–2025	Teaching Assistant, IIT Kharagpur and IIT Roorkee: Multimodal Urban Transport (NPTEL); Intersection Design (CEN 662); Engineering Drawing (CE13001). Laboratory sessions, simulation tools and geospatial analysis.

PROFESSIONAL SERVICE

Reviewer	<i>Transport Policy; Transportation Research Part A: Policy and Practice; Transportation Research Record; Travel Behaviour and Society; Journal of Transport & Health; Research in Transportation Business & Management; Expert Systems with Applications; PLOS Climate</i>
2025	Technical Session Chair, EASTS Conference, Jakarta
2023, 2024	Overall Coordinator, Annual Conference on Infrastructure (IBSR), IIT Kharagpur
2022–2023	Departmental Social Media Head, IIT Kharagpur
2020–2021	Placement Representative, IIT Roorkee
Member	ASCE Transportation & Development Institute; Transportation Research Group of India; World Conference on Transport Research Society; International Association for Travel Behaviour Research

RESEARCH SOFTWARE

DRUM	Dynamic Routing for Urban Mobility : pollution-aware route planning for commuters. React, Python, GraphHopper. Basis of J8 and the patent.
Survey	Stated-preference survey platform : adaptive choice experiments with live response analysis. React, Node.js, MongoDB.
PM _{2.5}	Next-day PM_{2.5} forecasting : machine-learning forecasts with a global viewer. Python, scikit-learn.
AQI	India air-quality dashboard : city-level analysis of real-time air quality. Streamlit, Plotly.
Mode share	Cycling and motorcycle mode-share dashboard : mode share in cities worldwide, with prediction. JavaScript.
Districts	Bharat Districts Explorer : district-level demographic and spatial data for India. D3.js.

METHODS AND TOOLS

Modelling	Discrete choice (hybrid latent class, ICLV, mixed logit, nested and inverse product-differentiation logit); panel econometrics; stated-preference design; willingness-to-pay estimation; MCDM
Learning	Deep learning; interpretable machine learning; geostatistics and kriging; LSTM and Prophet; computer vision for the built environment; convex optimisation
Software	Python, R, JavaScript, React, Node.js, SQL, MongoDB, Git, LaTeX; NLOGIT, SPSS, VISSIM, QGIS, Tableau, D3.js
Fieldwork	Low-cost sensor deployment; on-road exposure measurement by travel mode; two-wave panel survey administration

REFERENCES

Prof. Arkopal Kishore Goswami (Ph.D. advisor)

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Prof. Amit Agarwal (M.Tech advisor)

Associate Professor and Associate Dean, Corporate Interaction, IIT Roorkee
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Prof. Youngseo Kim (Postdoctoral advisor)

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